

## Proposition 2:

If  $\text{rank}(\Lambda_r)$  remains unchanged in crisis, but (i)  $\sigma^2$

$\lambda_{k,r}$  rises for one or more factors, and (ii)  $\lambda_{k,r}$

*is non-negligible and similar across currency for each global factor, i.e. react similarly to each global factor and the response is non-trivial, which means there is non-zero convergence in response to the global factors across currencies (autonomy compression, each currency has*

*less autonomy from global factors, that is, the common stochastic trends)* [this italicized part, what does it say? Yes, it says cointegration], **then** long-run correlations, that is the correlation of long-run currency returns across currencies increases, and factor volatility explains a larger share of the total variance of the long horizon returns for each currency.

By definition, cointegration is several other things, including:

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*.... $\lambda_{k,r}$  is non-negligible and similar across currency for each global factor, i.e. react similarly to each global factor and the response is non-trivial, which means there is non-zero convergence in response to the global factors across currencies (autonomy compression, each currency has less autonomy from global factors, that is, the common stochastic trends)*  
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So, if you remove some of the fluff there, this boils down to something we have exchanged back and forth: **If** Cointegration, **then** Strong Long Horizon Returns Correlations with High magnitude  $\rightarrow 1$ . This statement is the same as saying the *Contrapositive*: **If** Weak Long Horizon Returns Correlation with Low Magnitude  $\rightarrow 1$ , **then** No Cointegration. And, as you know, No Cointegration means: 1) Negation of the italicized, meaning those currencies do not react similarly or 2) Increased autonomy, or...

The Covid and Dot Com are more consistent with the contrapositive: weak correlation, no cointegration (0/4). The GFC less so, but you can argue GFC is also consistent because a) correlation is also weak and b) although you have cointegration, it is just 1 out of 4 max.

If you're still unclear about something, you can say it.